

A Model for Detecting Credit Card Fraud

Dan Caverly
Michael Hicks
Matthew Drummond
Sean Stack

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SUMMARY

Credit card fraud is a major challenge in today's economy, costing businesses and consumers billions of dollars annually. Traditional rule-based fraud detection methods are too rigid to keep pace with constantly evolving fraudulent schemes. This project proposed a machine learning solution to detect fraudulent transactions more effectively.

Utilizing publicly available IEEE-CIS Fraud Detection datasets from Kaggle and other sources, this project will explore and model transaction data to identify patterns distinguishing fraudulent activity from legitimate transactions.

Success of this model will be measured by achieving strong predictive performance, will recall ≥ 0.86 and precision ≥ 0.90 . Failure will be defined as an inability to outperform baseline models in terms of recall and precision.

INTRODUCTION

Financial institutions face a constant balancing act between managing risk and maintaining high customer satisfaction. One area where this act can raise tensions is in the handling of determining credit card fraud. Imagine a customer of a credit card company makes a purchase in a foreign country. The credit card company sees this as a suspicious transaction and places a block on the card to prevent further alleged fraudulent activity. However, the transaction was in fact initiated by the intended cardholder. Though the system did prevent the card holder from losing money, one would assume that this customer would be more than displeased with this inconvenience.

With a model that would understand the card holder's patterns, either travelling patterns or the types of items purchased at a certain time of year, the amount spent etc. this could prevent credit cards from being blocked when in fact the card holder is the one using the card and being able to identify true fraudulent transactions.

BACKGROUND

Financial fraud detection has become an increasingly urgent area of research as fraudulent credit/debit card transactions continue to rise worldwide. According to the Nilson Report, global card fraud losses totaled \$32.34 billion in 2021. This figure underscores both the scale of the problem and the challenge of combating it. Any financial losses burden financial institutions and

customers alike. Loss of capital for the common man limits his purchasing power, this can cause economic decline if consumer spending declines. For the financial institution, loss due to fraud can result in higher fees, lower interest rates on deposits and higher interest rates on lending.

Credit and debit card fraud takes many forms, including stolen cards, counterfeit transactions, account takeovers, online manipulation schemes, and many more. This diversity of scams highlights the inadequacy of traditional rule-based detection systems, which rely on fixed thresholds to determine when fraud may be occurring. Rather than relying on increased amounts of data to determine when fraudulent activity is occurring, companies continue to opt with traditional algorithms.

While simple and transparent, the traditional algorithms of fraud detection must be constantly updated to reflect new fraud tactics. Furthermore, these methods generate excessive amounts of false positives, which inconvenience customers. Our mission in this project is to use a wide range of data inputs to develop a fraud detection algorithm that features high accuracy/precision and low amounts of false positives.

PROPOSED TASKS/PLAN OF WORK

Task 1: Literature Review and Data Definition

- Research financial regulations, current credit card fraud issues/policies and relevant predictive modeling methods
- Identify variables influencing card block decision
 - Transaction amount
 - Purchase frequency
 - Location
 - Purchase channel (In-store, online, over the phone)
 - Customer tenure
 - Prior fraud history

Task 2: Dataset Assembly/Cleaning

- Review publicly accessible datasets, find ones suitable for this project
- If sufficient data cannot be located, create a simulation dataset modeled on realistic credit card characteristics.
- Include binary outcomes (“fraud” or “not fraud”) and continuous outcomes (card block duration in days)

Task 3: Model Development

Task 4: Evaluation

- Evaluate model performance based on accuracy, precision and mitigation of loss
 - Make necessary adjustments based on performance
 - Limit false positives

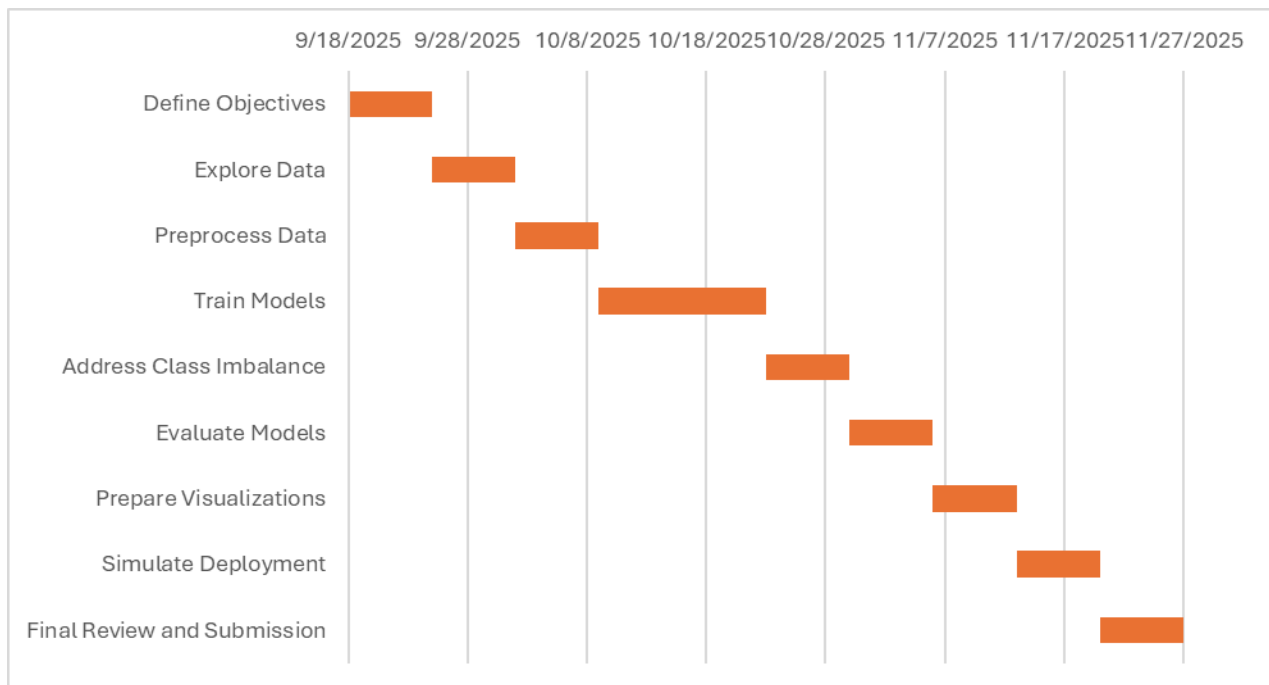
Task 5: Data Analysis and Visualization

- Use R packages such as tidyverse and ggplot2 to summarize and visualize the data
 - Identify present patterns
 - Detect outliers

Task 6: Present Findings

- Draft a written report describing the model and its accuracy
 - Create a call to action, suggestions for how financial institutions could adopt the model

SCHEDULE



QUALIFICATIONS

Project Leader: **Dan Caverly**

- Current employee of a financial institution
- Prior experience in R Studio
- Engineering Science student at the University of Buffalo

Data Architect: **Matthew Drummond**

- Familiar with R libraries such as ggplot2 and dplyr
- Python Programming
- Engineering Science student at the University of Buffalo

Data Scientist: **Michael Hicks**

- 2+ years experiences collecting and analyzing data in Manufacturing industry.
- Prior experience with R Studio
- Engineering Science student at the University of Buffalo

Scribe: **Sean Stack**

- Engineering Science student at the University of Buffalo

CONCLUSION

This project proposal outlines a data science project to improve credit card fraud detection through machine learning algorithms. By analyzing publicly available datasets and applying advanced classification techniques, this project aims to deliver a deployable model capable of detecting fraudulent transactions with high accuracy and precision with low amounts of false positives. Successful completion of this project will provide actionable insights for financial institutions along with a practical solution to one of the most complex issues in modern finance. Furthermore, the potential success and application of this model could assist models for similar situations.

REFERENCES

Credit Card and Debit Card Fraud. Office of the Comptroller of the Currency. (2024, December 11). <https://www.occ.gov/topics/consumers-and-communities/consumer-protection/fraud-resources/credit-card-and-debit-card-fraud.html>

Card Fraud Losses Worldwide - 2021. Nilson Report. (2024, November 6). <https://nilsonreport.com/articles/card-fraud-losses-worldwide/>

Rules-Based Fraud Detection Definition - FraudNet. FraudNet. (n.d.). <https://www.fraud.net/glossary/rules-based-fraud-detection#what-is-rules-based-fraud-detection>